

Computer Task 1 – How to interpret finite element results

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1. Introduction

In this computer task, the aim and objective is to carry out a series of numerical simulations using the Finite Element Methods for the analysis of different mechanical components. The focus of this task is on validating the correctness and quality of numerical solutions obtained by FEM (FE solution) by comparing them with existing analytical solutions.

By taking into consideration various mechanical components like Continuum beam, infinite thick pipe and flat bar with edge notches in this computer task, this report evaluates modelling approaches, mesh dependence, and different stress, displacement computation methods in ANSYS mechanical.

1.1 Material Description:

In all of the subtask in this report, the material is taken to be isotropic and linearly elastic. The mechanical property is assigned to the models are as follows:

- Young's Modulus (E): 2×10^{11} Pa
- Poisson's Ratio (ν): 0.3

2. Subtask 1: Continuum beam – tip loaded

2.1 Description of the problem

This subtask covers the finite element analysis of a beam subjected to bending. Numerical results are compared to the analytical solution as described in the theory part, especially in relation to the displacement at the tip of the beam.

The report includes three graphs showing the vertical displacement, bending stress, and shear stress along the beam's upper surface. In all cases, the analytical solution is shown in conjunction with the numerical solution to allow a comparison. Additionally, the full von Mises stress distribution is shown for the two different load cases.

Fig. 1.

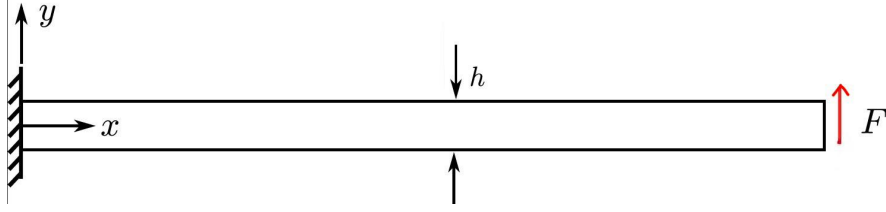


Figure 1: An end-loaded cantilever beam with height h and width b . A load F acts on the right end of the beam.

2.2 Analytical Solution

The analytical verification of the cantilever beam is based on the Euler-Bernoulli beam theory, supplemented by the Timoshenko shear deformation correction to account for the beam's depth. As per ref. [1], the relationship between the bending moment $M(x)$ and the vertical displacement $v(x)$ is governed by the moment-curvature equation:

$$EI \frac{d^2 v}{dx^2} = -M(x) \quad (1)$$

For a beam of length L with a concentrated load P at the free end ($x = L$), the internal moment at any section x is $M(x) = P(L - x)$. Substituting this into the governing equation yields:

$$EI \frac{d^2 v}{dx^2} = -P(L - x) = P(x - L) \quad (2)$$

Integrating Equation (2) once gives the slope $\theta(x)$:

$$EI \frac{dv}{dx} = \frac{Px^2}{2} - PLx + C_1 \quad (3)$$

Integrating a second time gives the displacement $v(x)$:

$$EI v(x) = \frac{Px^3}{6} - \frac{PLx^2}{2} + C_1x + C_2 \quad (4)$$

Applying the boundary conditions for a perfectly clamped support at the root ($v(0) = 0$ and $v'(0) = 0$), the constants of integration are found to be $C_1 = 0$ and $C_2 = 0$. The maximum displacement due to bending at the tip ($x = L$) is:

$$\delta_{bending} = \frac{PL^3}{3EI} \quad (5)$$

2.2.1 Shear Deflection Correction

For beams where the height-to-length ratio is significant, shear deformation must be included. The additional deflection due to shear is defined as:

$$\delta_{shear} = \frac{PL}{kAG} \quad (6)$$

where G is the shear modulus, A is the cross-sectional area, and k is the shear correction factor (taken as $5/6$ for rectangular sections). The total analytical displacement is the sum of both components: $\delta_{total} = \delta_{bending} + \delta_{shear}$.

2.2.2 Geometric and Material Parameters

The geometric and material properties used for the analytical evaluation are listed below:

- **Length (L):** 100 mm
- **Applied Load (P):** 1000 N
- **Young's Modulus (E):** 200×10^3 MPa
- **Poisson's Ratio (ν):** 0.3
- **Width (b):** 5 mm
- **Height (h):** 20 mm

2.2.3 Numerical Parameters and Evaluation

. Using these parameters, the derived properties are:

- **Area:** $A = b \cdot h = 100 \text{ mm}^2$
- **Moment of Inertia:** $I = \frac{bh^3}{12} = 3333.33 \text{ mm}^4$
- **Shear Modulus:** $G = \frac{E}{2(1+\nu)} = 76923.1 \text{ MPa}$

Displacement Results: Applying the values to Equations (5) and (6):

$$\delta_{bending} = \frac{1000 \cdot 100^3}{3 \cdot 200000 \cdot 3333.33} = 0.5000 \text{ mm} \quad (7)$$

$$\delta_{shear} = \frac{1000 \cdot 100}{(5/6) \cdot 100 \cdot 76923.1} = 0.0156 \text{ mm} \quad (8)$$

$$\delta_{total} = 0.5156 \text{ mm} \quad (9)$$

Stress Results: The maximum bending stress (σ_{max}) at the clamped end ($y = h/2$) and average shear stress (τ_{avg}) are:

$$\sigma_{max} = \frac{M_{max} \cdot (d/2)}{I} = \frac{(1000 \cdot 100) \cdot 10}{3333.33} = 300.0 \text{ MPa} \quad (10)$$

$$\tau_{xy} = 0 \text{ MPa} \quad (11)$$

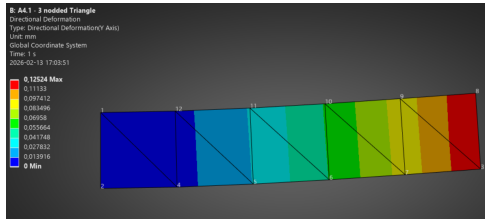
The mean shear stress in the beam can be written as

$$\bar{\tau}_{xy}(x, y) = \frac{6T(x)}{bd^3} \left(\frac{d^2}{4} - y^2 \right) \quad (12)$$

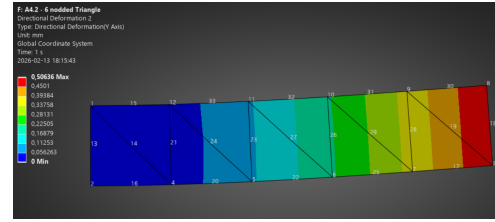
In the analytical solution, the shear stress τ_{xy} is zero at the extreme edge ($y = \pm d/2$). This is because these are traction-free horizontal surfaces where no external tangential loads are applied.

2.3 FE results

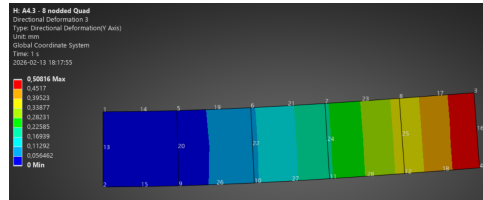
Exercise A4.1



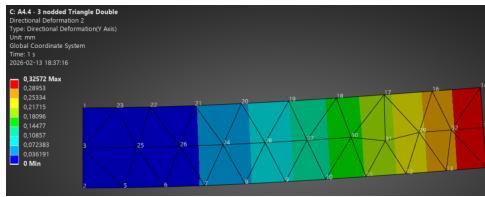
(a) 3 Nodded Triangle



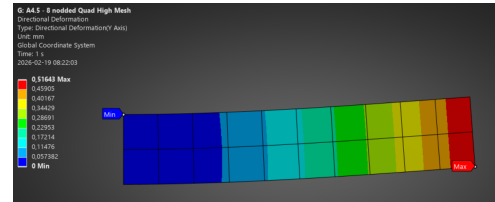
(b) 6 Nodded triangle



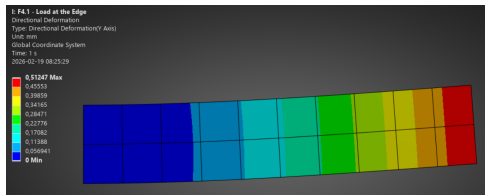
(c) 8 Nodded Quadrilateral



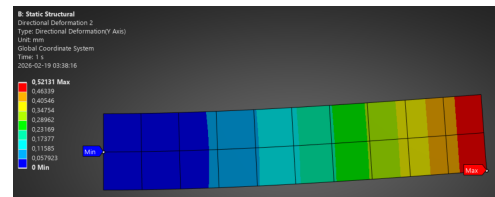
(d) 3 noded Triangle with High Mesh



(e) 8 Nodded Quadrilateral with High Mesh



(f) 8 Nodded Quadrilateral with Load along the Edge



(g) 8 Nodded Quadrilateral with Engineer's Thoery of Bending Support Condition

Figure 2: Vertical Displacement: ANSYS Simulation Results.

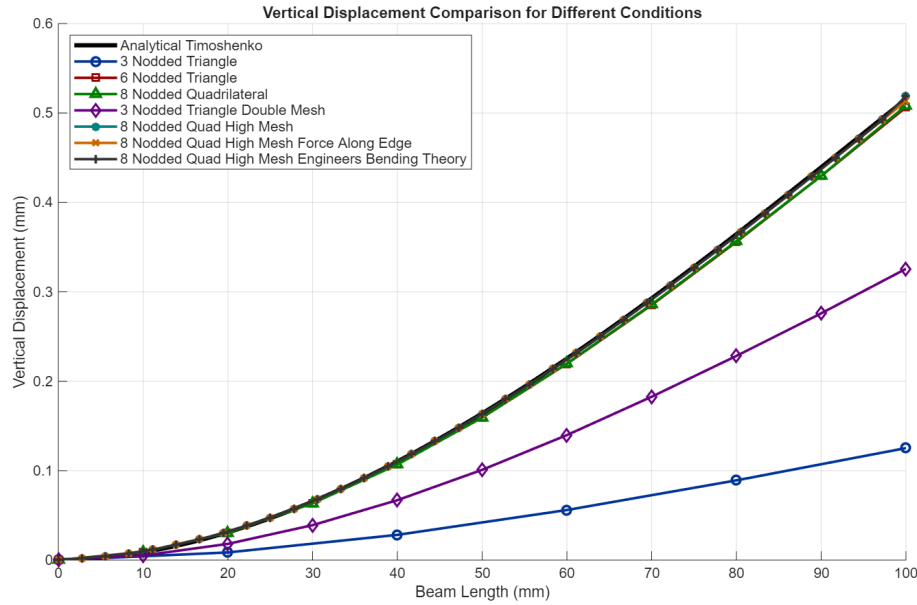


Figure 3: Vertical displacement along the top surface compared with analytical solution.

Results:

As shown in the figure 3, the vertical displacement results show difference in relation to the type of elements used in the analysis.

For instance, the 8-nodded quadrilateral (both high & low mesh) elements (A4.3, A4.5, F4.1 & F4.2) as well as 6-nodded elements (A4.2) display a high degree of concurrence with the analytical solution, approaching a displacement value of 0.51 mm at the tip of the beam.

On the other hand, the 3-nodded triangular mesh (both high & low mesh) display a result which is significantly less than the analytical solution.

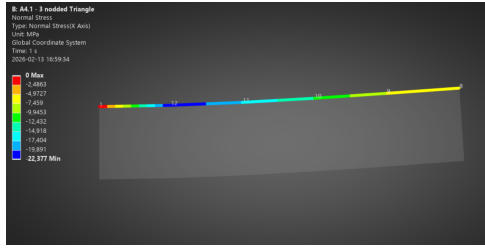
Comments:

Based on the plot obtained, the FE solutions with higher order elements give nearest value to the analytical solution.

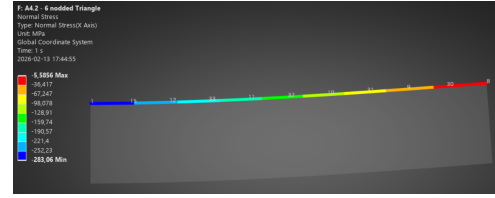
In the case of F4.1, even though the load applied is smeared along the edge of the beam, there is negligible effect on the global tip displacement.

In the case of F4.2, as described in the Engineer's Theory of Bending, affects the local stress state in this region. This is because transverse strains are allowed, but transverse stresses are suppressed. This causes a reduction in the stiffness contribution from this fixed end, thereby increasing the deflection to the desired value.

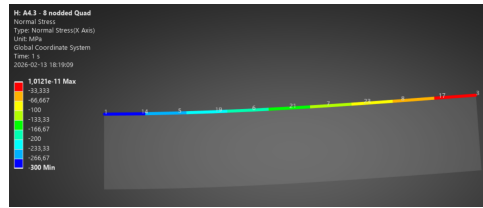
Bending Stress



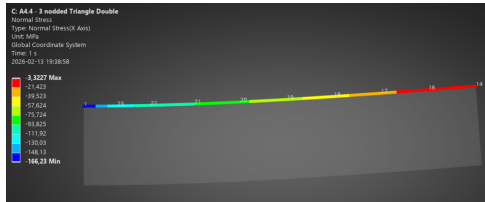
(a) 3 Nodded Triangle



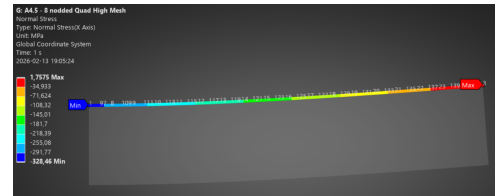
(b) 6 Nodded triangle



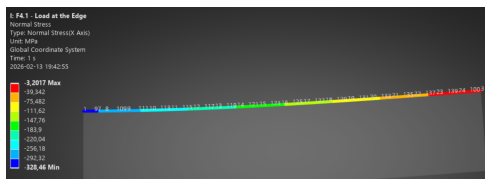
(c) 8 Nodded Quadrilateral



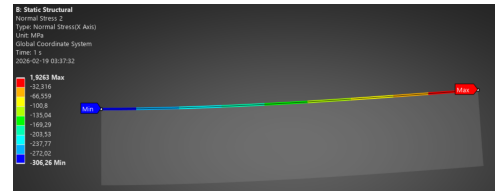
(d) 3 nodded Triangle with High Mesh



(e) 8 Nodded Quadrilateral with High Mesh



(f) 8 Nodded Quadrilateral with Load along the Edge



(g) 8 Nodded Quadrilateral with Engineer's Thoery of Bending Support Condition

Figure 4: Bending Stress: ANSYS Simulation Results.

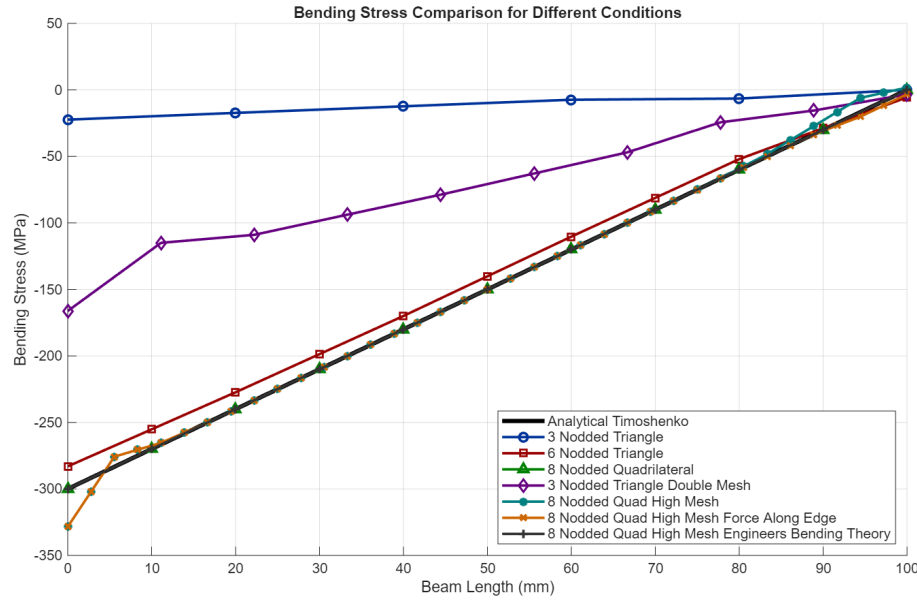


Figure 5: Bending Stress along the top surface compared with analytical solution.

Results:

As shown in the figure 5, the bending stress result exhibits variation with respect to the type of element used in the analysis.

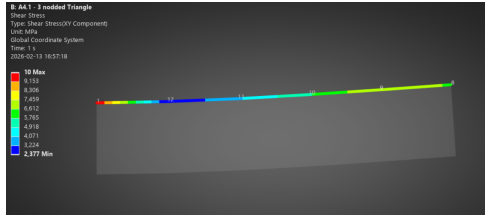
For instance, in the higher order element like 8-nodded quadrilateral and 6-nodded triangle mesh elements display close concurrence with the analytical solution. The bending stress distribution in the top surface follows a linear trend. In case of 3-node triangle, the plot is pretty flat line rather a linear trend. Even with the refined mesh, the stress values are significantly lower than the analytical maximum at the fixed support, failing to capture the peak magnitude.

In both F4.1 & F4.2 cases, the stress distribution at the support or along the mid span dose not have any noticeable change or difference compared to the analytical solution. Even in the F4.2 case, releasing the transverse constraints at the support , resulting in a distribution that closely matches the Engineer's theory of bending.

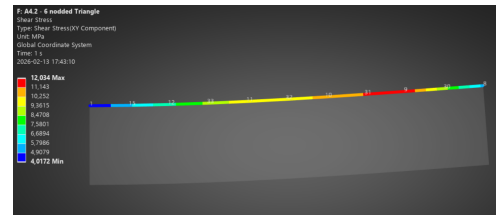
Comments:

From the plot, the linear triangular elements are unsuitable for stress analysis in bending problems. The higher order elements like 6-nodded Triangular or 8-nodded quadrilateral is required to capture the linear variation of the bending stress along the length efficiently.

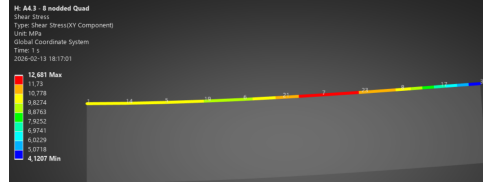
Shear Stress



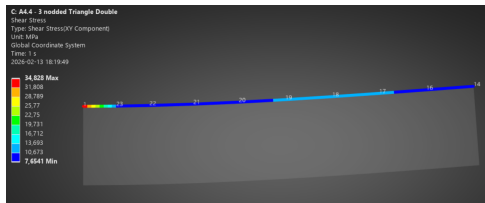
(a) 3 Nodded Triangle



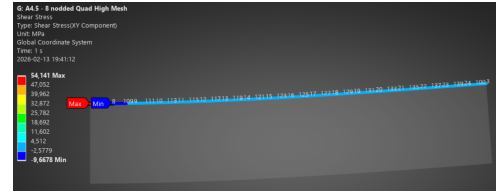
(b) 6 Nodded triangle



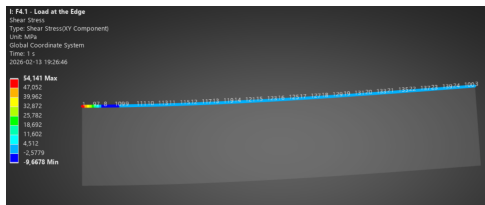
(c) 8 Nodded Quadrilateral



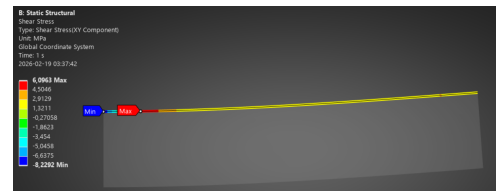
(d) 3 nodded Triangle with High Mesh



(e) 8 Nodded Quadrilateral with High Mesh



(f) 8 Nodded Quadrilateral with Load along the Edge



(g) 8 Nodded Quadrilateral with Engineer's Thoery of Bending Support Condition

Figure 6: Shear Stress: ANSYS Simulation Results.

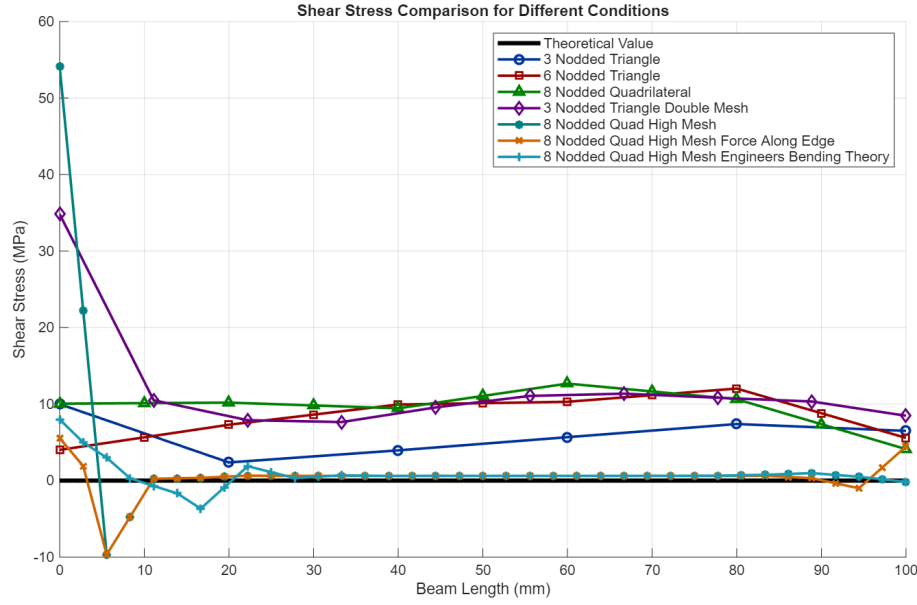


Figure 7: Shear Stress along the top surface compared with analytical solution.

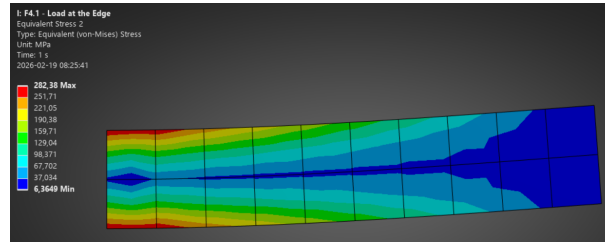
Results:

As shown in the figure 7, the linear as well as quadratic triangular mesh elements and quadratic quadrilateral with lesser mesh fail to satisfy the zero-stress boundary condition, as it displays large, erratic fluctuations along the top surface due to small number node to converge. Analytically, the shear stress on the top surface of the beam should be zero. As compared to the linear triangular elements, the quadratic triangular and 8-nodded quadrilateral with lesser mesh element ; 8-nodded Quadrilateral with higher mesh element produce more smoother result that stays close to zero, approximately capturing the traction-free boundary condition.

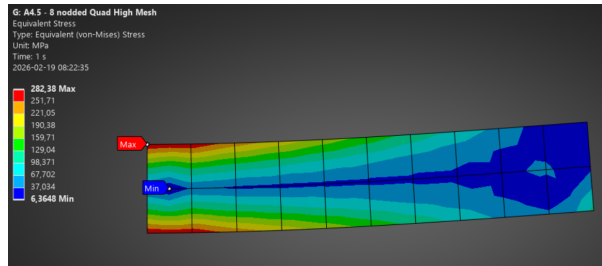
Comments:

With higher order elements with higher mesh , the FE solution for the shear stress converges towards the analytical value (zero) compare to the linear triangular element or higher order element with lesser mesh.

F4.1 : Von Mises Stress



(a) Distributed Load (Smeared across the edge)



(b) Point Load (Single Node)

Figure 8: Von Mises : ANSYS Simulation

Comments:

As in the figure 8, in the case of point-load model, the Von Mises stress plot is dominated by a severe and concentrated peak at the single node where the force is applied.

In the case of smeared load model, by distributing the load across the entire vertical face of the tip, the localized peak is eliminated. Instead, a uniform stress is observed at the free end.

3. Subtask 2: Infinite thick pipe

3.1 Description of the problem

This subtask deals with the stress and displacement analysis of an infinite thick pipe (2D) under various element 2D behaviour such as plane stress, plane strain, and axisymmetric conditions. This analysis aims to determine the influence of 2d behaviour on the radial stress, tangential stress, and radial displacement.

The report contains three graphs for comparison purposes: one for radial stress, one for tangential stress, and one for radial displacement. Both numerical and analytical results are included for each graph to determine the level of agreement and to discuss the physical validity of the modeling assumptions.

Fig. 9.

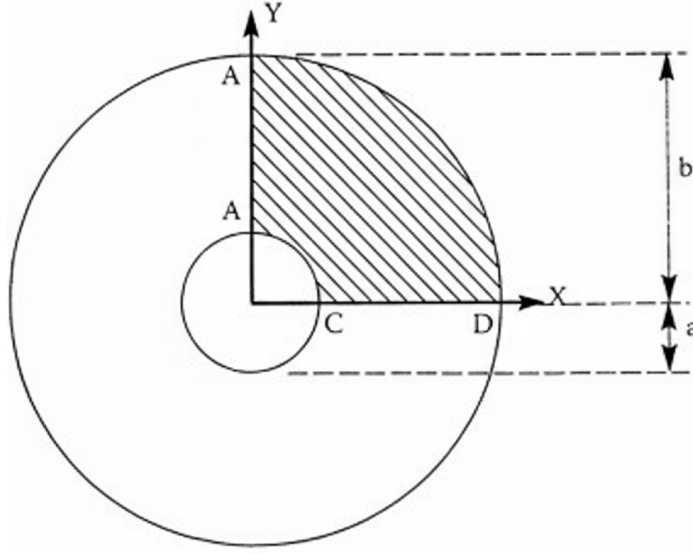


Figure 9: An infinte thick pipe (2D) with inner radius a and outer radius b . A load p acts on the inner radius.

3.2 Analytical solution

The analytical solution for a infinte thick pipe (2D) subjected to internal pressure is derived from the theory of elasticity, specifically Lamé's equations as presented by Timoshenko and Goodier. The radial stress (σ_r) and hoop (tangential) stress (σ_θ) at any radial distance r are given by:

$$\sigma_r(r) = \frac{pa^2}{b^2 - a^2} \left(1 - \frac{b^2}{r^2} \right) \quad (13)$$

$$\sigma_\theta(r) = \frac{pa^2}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right) \quad (14)$$

where:

- a is the internal radius of the pipe.
- b is the external radius of the pipe.
- p is the internal applied pressure.
- r is the radial distance from the center ($a \leq r \leq b$).

Depending on the modeling assumptions for the 2D slice, the axial stress (σ_z) is defined as:

$$\sigma_z = \begin{cases} 0 & \text{for Plane Stress} \\ \nu(\sigma_r + \sigma_\theta) & \text{for Plane Strain} \end{cases} \quad (15)$$

The radial displacement $u(r)$ is expressed as:

$$u(r) = \frac{r}{E} (\sigma_\theta - \nu(\sigma_r + \sigma_z)) \quad (16)$$

3.2.1 Geometric and Material Parameters

The geometric and material properties used for the analytical evaluation are listed below:

- **Inner Radius (a):** 3 mm
- **Outer Radius (b):** 9 mm
- **Uniform Internal Pressure (P):** 1 MPa
- **Young's Modulus (E):** 200×10^3 MPa
- **Poisson's Ratio (ν):** 0.3
- **Thickness of 2D Slice (t):** 1 mm

3.2.2 Numerical Evaluation

The parameters for this subtask are $p = 1$ MPa, $a = 3$ mm, $b = 9$ mm, $E = 200 \times 10^3$ MPa, and $\nu = 0.3$. The geometric constant $C = \frac{pa^2}{b^2 - a^2}$ is calculated as:

$$C = \frac{1 \cdot 3^2}{9^2 - 3^2} = \frac{9}{81 - 9} = \frac{9}{72} = 0.125 \text{ MPa} \quad (17)$$

At the inner radius ($r = a = 3$ mm):

$$\sigma_r(3) = 0.125 \left(1 - \frac{9^2}{3^2} \right) = 0.125(1 - 9) = -1.0 \text{ MPa} \quad (18)$$

$$\sigma_\theta(3) = 0.125 \left(1 + \frac{9^2}{3^2} \right) = 0.125(1 + 9) = 1.25 \text{ MPa} \quad (19)$$

At the outer radius ($r = b = 9$ mm):

$$\sigma_r(9) = 0.125 \left(1 - \frac{9^2}{9^2} \right) = 0 \text{ MPa} \quad (20)$$

$$\sigma_\theta(9) = 0.125 \left(1 + \frac{9^2}{9^2} \right) = 0.25 \text{ MPa} \quad (21)$$

Axial Stress (Plane Stress): For the plane stress condition, the axial stress is zero throughout the thickness:

$$\sigma_{axial} = 0 \quad (22)$$

Axial Stress (Plane Strain): For the plane strain condition, the axial stress is constant throughout the thickness:

$$\sigma_z = 0.3(-1.0 + 1.25) = 0.075 \text{ MPa} \quad (23)$$

Radial Displacement Calculation The radial displacement $u(r)$ depends on the modeling assumption due to the axial stress term σ_z .

Case 1: Plane Stress ($\sigma_z = 0$)

$$u(r) = \frac{r}{E}(\sigma_\theta - \nu\sigma_r) \quad (24)$$

At the inner radius ($r = 3$):

$$u(3) = \frac{3}{200000}(1.25 - 0.3(-1.0)) = \frac{3}{200000}(1.55) = 2.325 \times 10^{-5} \text{ mm} \quad (25)$$

At the outer radius ($r = 9$):

$$u(9) = \frac{9}{200000}(0.25 - 0.3(0)) = \frac{9}{200000}(0.25) = 1.125 \times 10^{-5} \text{ mm} \quad (26)$$

Case 2: Plane Strain ($\sigma_z = \nu(\sigma_r + \sigma_\theta)$) First, calculate the constant axial stress:

$$\sigma_z = 0.3(-1.0 + 1.25) = 0.3(0.25) = 0.075 \text{ MPa} \quad (27)$$

Now, calculate displacement using $u(r) = \frac{r}{E}[\sigma_\theta - \nu(\sigma_r + \sigma_z)]$: At the inner radius ($r = 3$):

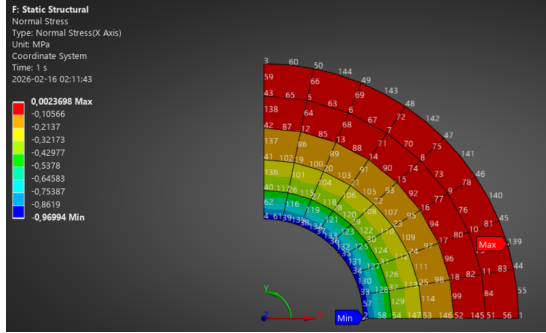
$$u(3) = \frac{3}{200000}[1.25 - 0.3(-1.0 + 0.075)] = \frac{3}{200000}(1.5275) = 2.291 \times 10^{-5} \text{ mm} \quad (28)$$

At the outer radius ($r = 9$):

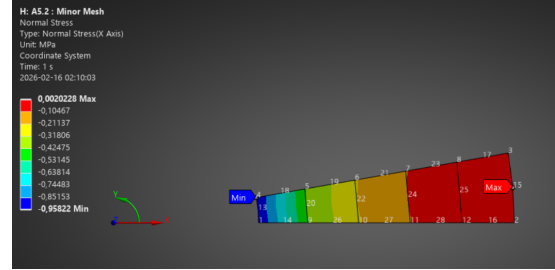
$$u(9) = \frac{9}{200000}[0.25 - 0.3(0 + 0.075)] = \frac{9}{200000}(0.2275) = 1.024 \times 10^{-5} \text{ mm} \quad (29)$$

3.3 FE results

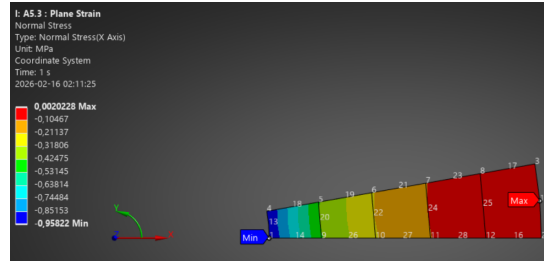
Radial Stress



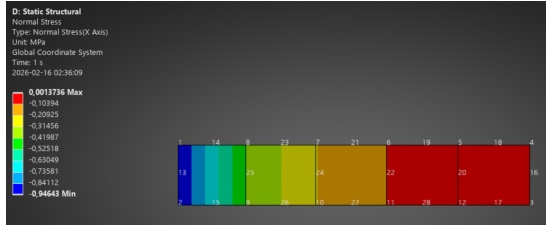
(a) 2D Plane Stress



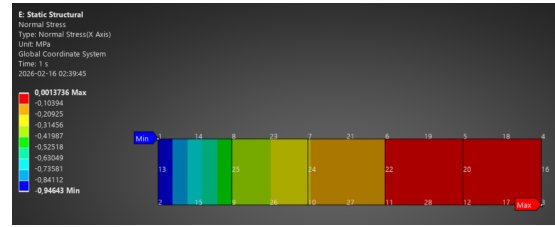
(b) 2D Plane Stress minor element



(c) 2D Plane Strain



(d) Axisymmetry for 2D Plane Stress



(e) Axisymmetry for 2D Plane Strain

Figure 10: Radial Stress: ANSYS Simulation Results.

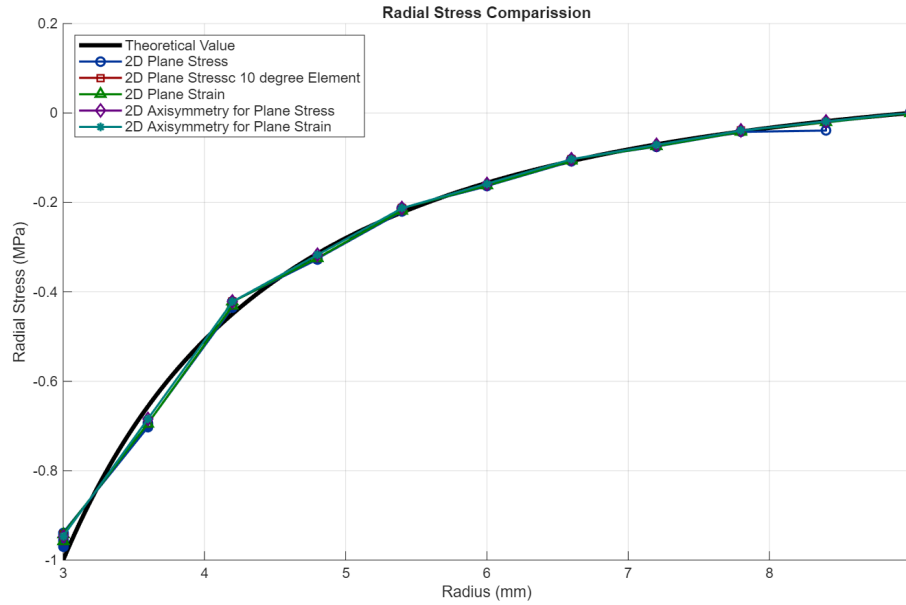


Figure 11: Radial Stress Comparison between analytical solution, 2D Plane Stress, 2D Plane Strain and Axisymmetry Condition.

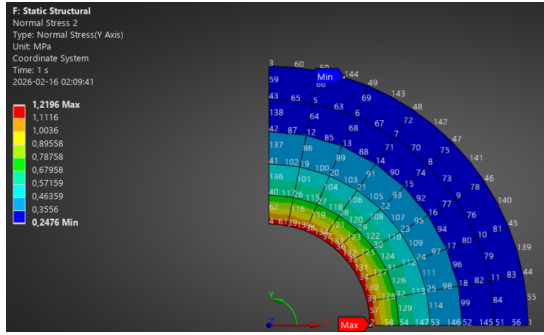
Results:

As shown in the Figure 11, the radial stress results across all three element 2D behaviour types are almost identical, follows same trend and align with the analytical solution.

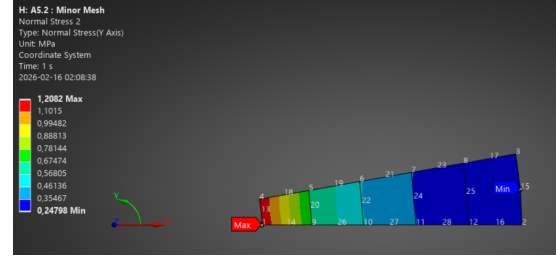
Comments:

As seen in the result, it is noted that irrespective of whether the 2D behaviour used is plane stress or plane strain, the radial stress is almost identical. This validates that all three 2D behaviours are providing consistent boundary traction.

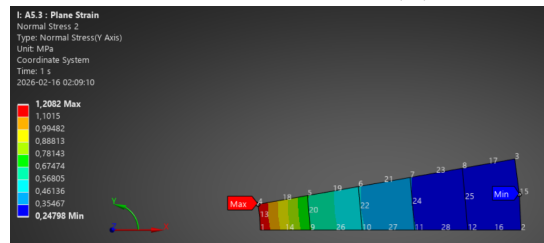
Hoop(Tangential) Stress



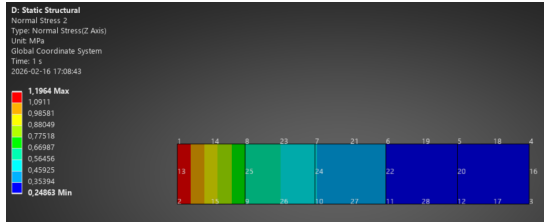
(a) 2D Plane Stress



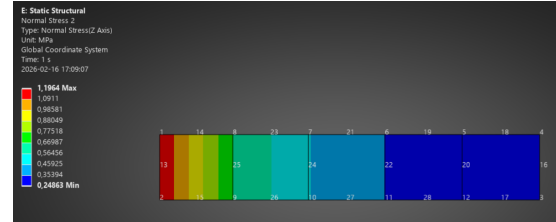
(b) 2D Plane Stress minor element



(c) 2D Plane Strain



(d) Axisymmetry for 2D Plane Stress



(e) Axisymmetry for 2D Plane Strain

Figure 12: Hoop Stress: ANSYS Simulation Results.

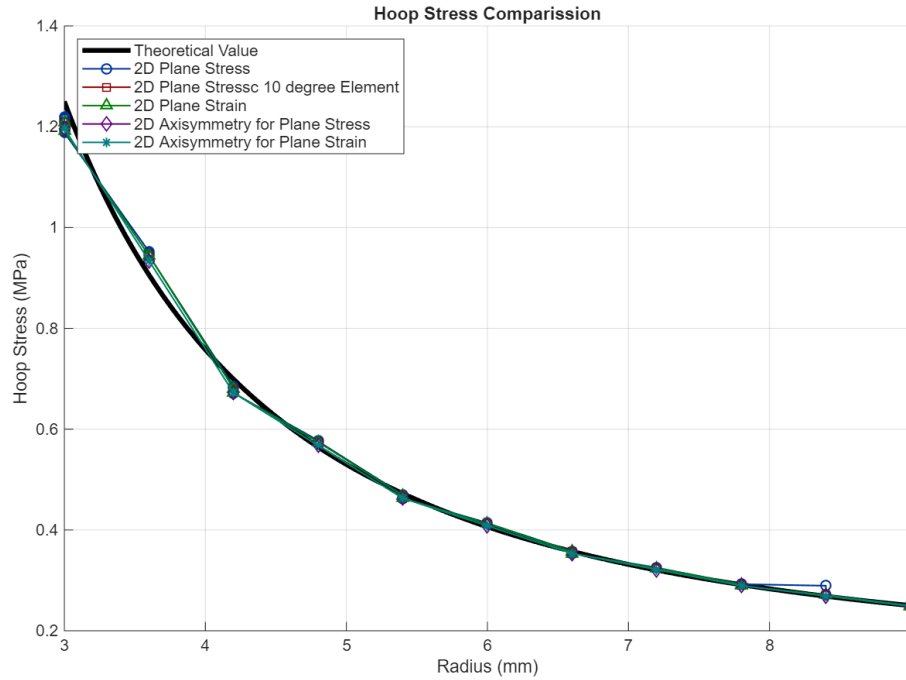


Figure 13: Hoop Stress Comparison between analytical solution, 2D Plane Stress, 2D Plane Strain and Axisymmetry Condition.

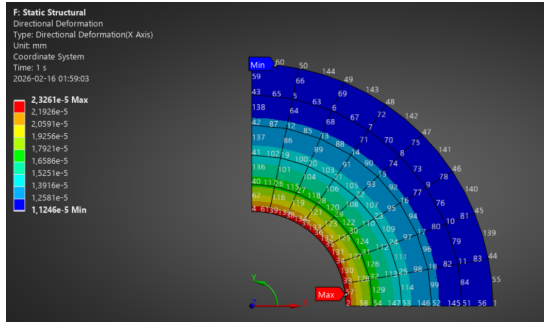
Results:

As shown in the Figure 13, the hoop stress results across all three 2D behaviour element types are almost identical, follows same trend and align with the analytical solution.

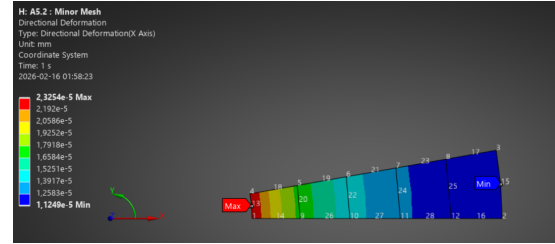
Comments:

As seen in the result, it is noted that irrespective of whether the 2D behaviour used is plane stress or plane strain, the hoop stress is almost identical.

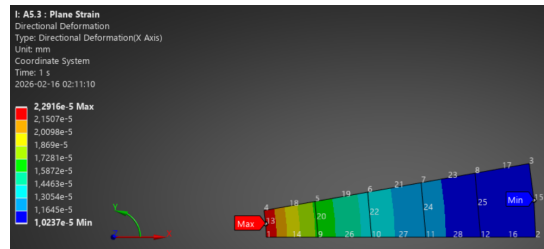
Radial Displacement



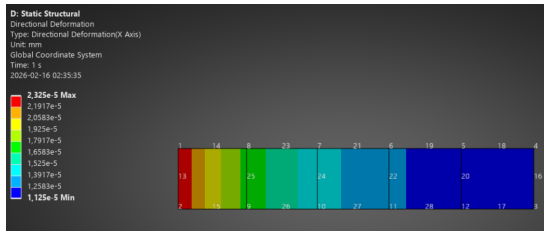
(a) 2D Plane Stress



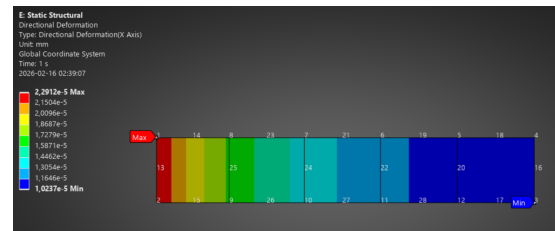
(b) 2D Plane Stress minor element



(c) 2D Plane Strain



(d) Axisymmetry for 2D Plane Stress



(e) Axisymmetry for 2D Plane Strain

Figure 14: Radial Displacement: ANSYS Simulation Results.

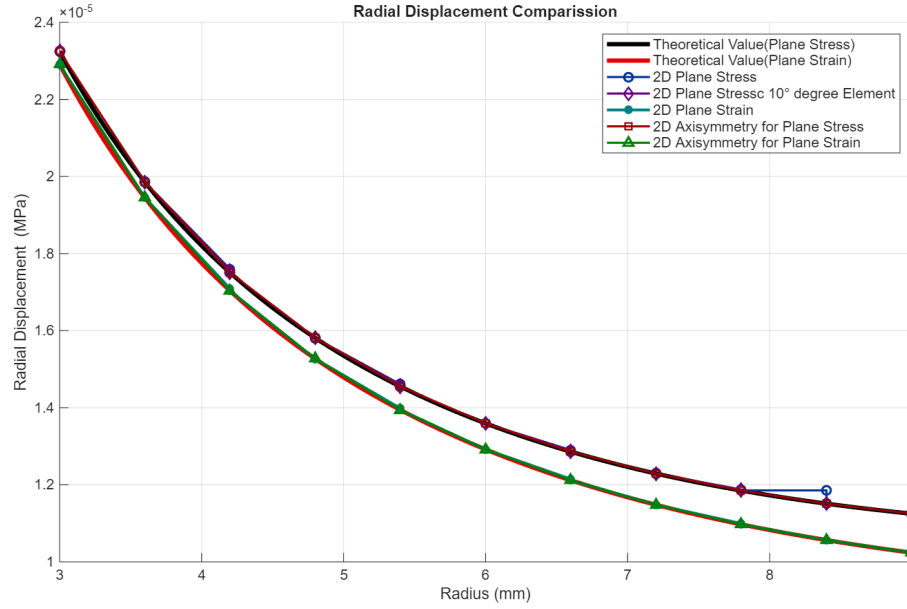


Figure 15: Radial Displacement Comparison between analytical solution, 2D Plane Stress, 2D Plane Strain and Axisymmetry Condition.

Results:

As shown in the Figure 15, the plane stress case yields a higher radial displacement compared to the plane strain and axisymmetric case.

Comments:

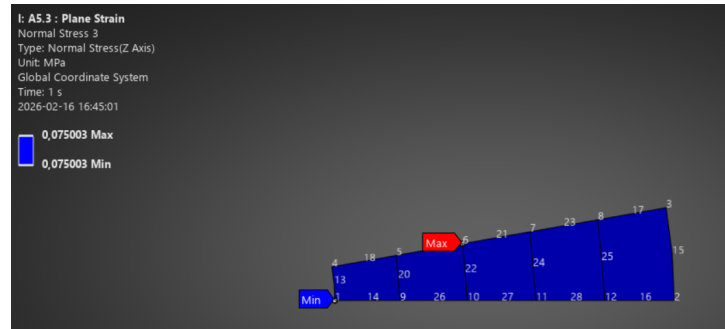
The difference in displacement can be explained by the Poisson Effect, together with the constraint in the z-axis direction.

Plane Stress: This is a element behaviour where the pipe is assumed to be a very thin object. The pipe is unconstrained in the z-axis direction and can contract in this direction as it expands in the radial direction.

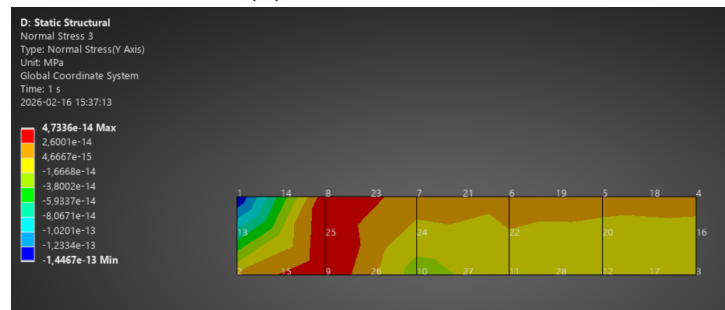
Plane Strain: This model is based on an “infinite” pipe with zero strain in the z-axis direction. To satisfy this condition, a positive stress in the z-axis direction, $\sigma_z = \nu(\sigma_r + \sigma_\theta)$, is required. This “stiffness” in the z-axis direction limits the expansion in the radial direction, resulting in a smaller displacement.

So, the physically correct result for an “infinite” pipe can be obtained from both Plane Strain and Axisymmetric analyses. Although Plane Stress could be a valid numerical result for a thin plate, it is a modeling error in a infinite thick pipe.

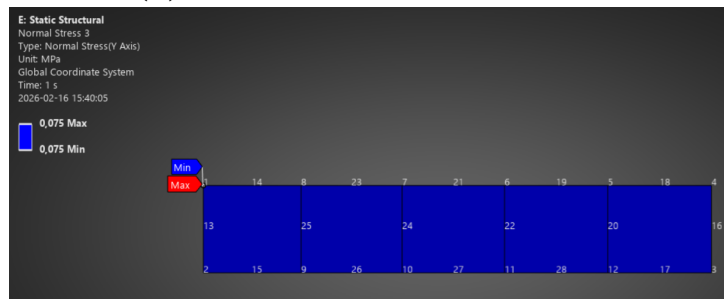
Axial Stress



(a) 2D Plane Strain



(b) Axisymmetry for 2D Plane Stress



(c) Axisymmetry for 2D Plane Strain

Figure 16: Axial Stress: ANSYS Simulation Results.

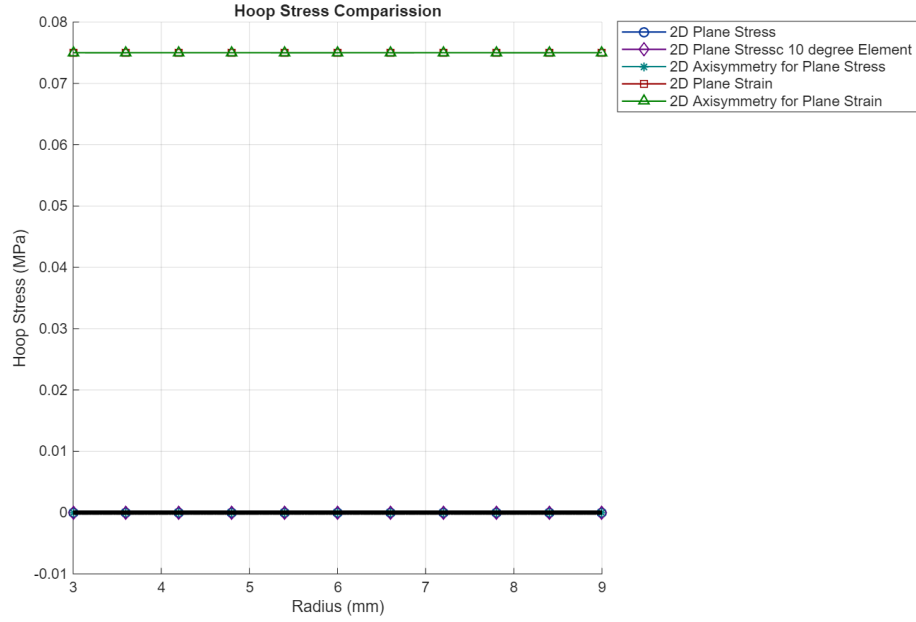


Figure 17: Axial Stress Comparison between 2D Plane Stress, 2D Plane Strain and Axisymmetry Condition.

Results:

In Plane Stress, the axial stress is exactly zero by definition. In Plane Strain and Axisymmetric models, a constant axial stress of approximately 0.075 MPa is observed across the thickness.

4. Subtask 3: Flat bar with edge notches

4.1 Description of the problem

This subtask involves the evaluation of the stress distribution in the notched edge component. It compares the numerical results to the reference results given in the handbook. The evaluation begins by using a coarse mapped mesh to display the stress plots for the averaged, unaveraged, and elemental mean stress representations of the normal stress σ_x . The differences between the representations of the stresses are analyzed.

And, a mesh convergence evaluation is performed using the automatic refinement. Stress contour plots are displayed for the refined meshes, along with a convergence plot showing the variation of the maximum stress with the solution number. This ensures the numerical results converge to a reliable answer.

Fig. 18.

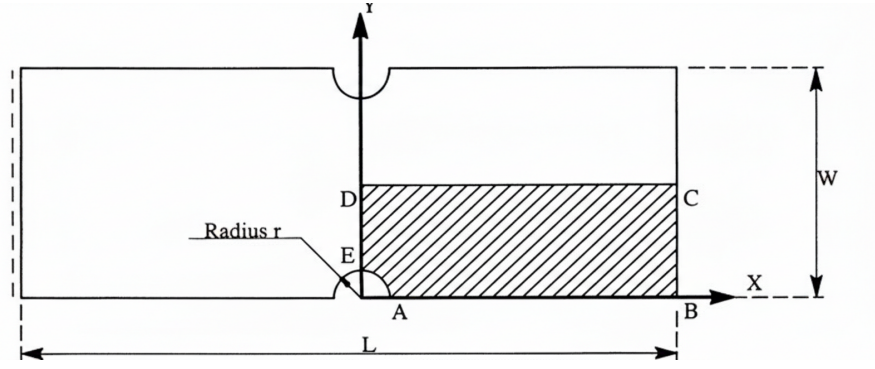


Figure 18: A flat bar with edge notches with notch radius r .

4.2 Analytical solution

The analytical verification for the plate with edge notches is based on the stress concentration factor found in the engineering handbooks. The analysis focuses on the maximum normal stress σ_{max} occurring at the root of the notch (Point E).

4.2.1 Geometry and Material Parameters

The geometric and material properties used for the analytical evaluation are listed below:

- **Total Width (B):** 80 mm (0.08 m)
- **Notch Radius (r):** 10 mm (0.01 m)
- **Thickness (t):** 1 mm (0.001 m)
- **Net Width (b):** $B - 2r = 60$ mm (0.06 m)
- **Young's Modulus (E):** 200×10^3 MPa
- **Poisson's Ratio (ν):** 0.3

4.2.2 Load Integration and Bending Moment

The plate is subjected to a linearly distributed pressure load $p(y)$ along the edge, defined as:

$$p(y) = 10^7 \left(1 - \frac{y}{0.04} \right) \text{ Pa} \quad (30)$$

To determine the nominal stress, we must first calculate the resulting bending moment M about the center of the plate ($y = 0.04$ m). The moment is obtained by integrating the pressure distribution multiplied by the moment arm ($0.04 - y$) and the thickness t over the loaded region ($0 \leq y \leq 0.04$):

$$M = \int_0^{0.04} p(y)(0.04 - y)t \, dy \quad (31)$$

Substituting the expression for $p(y)$:

$$M = \int_0^{0.04} \left[10^7 \left(\frac{0.04 - y}{0.04} \right) \right] (0.04 - y) t \, dy \quad (32)$$

$$M = \frac{10^7 t}{0.04} \int_0^{0.04} (0.04 - y)^2 \, dy \quad (33)$$

Performing the integration:

$$M = \frac{10^7(0.001)}{0.04} \left[-\frac{(0.04 - y)^3}{3} \right]_0^{0.04} \quad (34)$$

$$M = 2.5 \times 10^5 \left(0 - \left(-\frac{0.04^3}{3} \right) \right) = 2.5 \times 10^5 \cdot \frac{0.000064}{3} \quad (35)$$

$$M = 5.333 \text{ Nm} \quad (36)$$

4.2.3 Calculation of Nominal Stress

The nominal bending stress σ_{nom} at the root of the notch is calculated using the standard flexure formula for a rectangular section of net width b and thickness t :

$$\sigma_{nom} = \frac{6M}{tb^2} \quad (37)$$

Substituting the calculated moment and geometric values:

$$\sigma_{nom} = \frac{6 \cdot 5.333}{0.001 \cdot (0.06)^2} = \frac{32.0}{0.0000036} = 8.89 \times 10^6 \text{ Pa} = 8.89 \text{ MPa} \quad (38)$$

4.2.4 Stress Concentration Factor (K_t)

The maximum stress is related to the nominal stress by the stress concentration factor K_t :

$$\sigma_{max} = K_t \sigma_{nom} \quad (39)$$

The factor K_t is determined from handbook charts for a flat bar with opposite U-notches in bending, based on the following geometric ratios:

- **Notch-to-Width Ratio:** $r/b = 10/60 \approx 0.167$
- **Width Ratio:** $B/b = 80/60 \approx 1.33$

Using these ratios with the standard stress concentration curves, the value of K_t is estimated to be approximately 1.75.

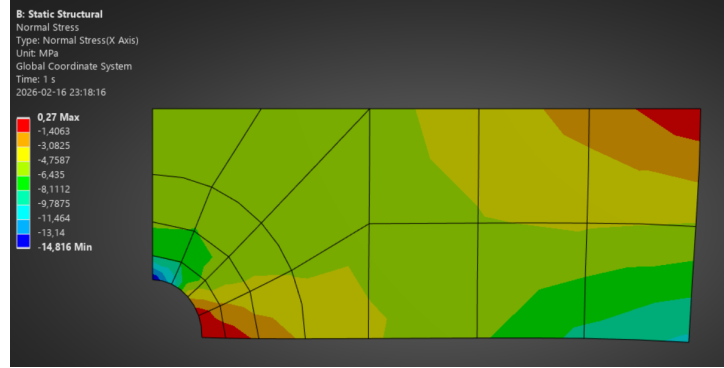
4.2.5 Maximum Theoretical Stress

Finally, the analytical maximum stress is:

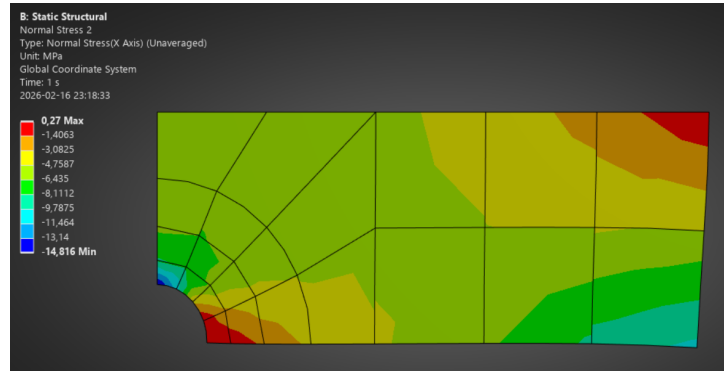
$$\sigma_{max} = 1.75 \cdot 8.89 \text{ MPa} \approx 15.56 \text{ MPa} \quad (40)$$

4.3 FE results

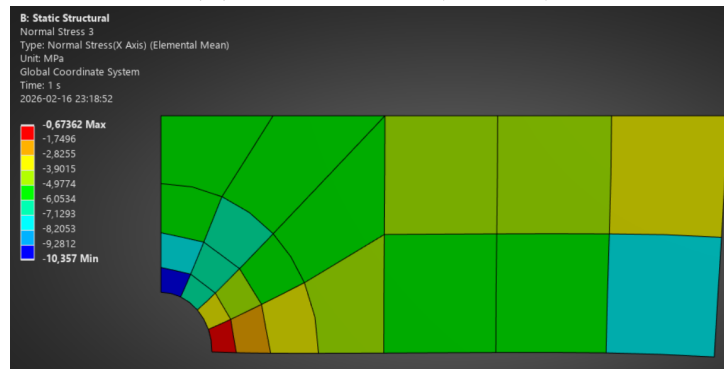
Normal Stress



(a) Normal Stress (*Avg*)

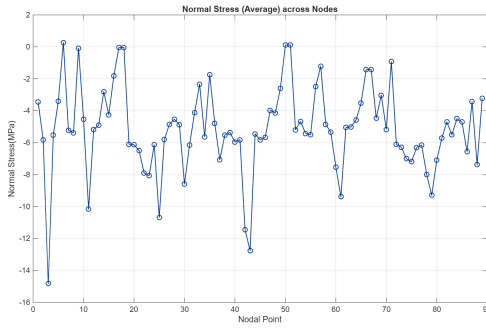


(b) Normal Stress (*Un-Avg*)

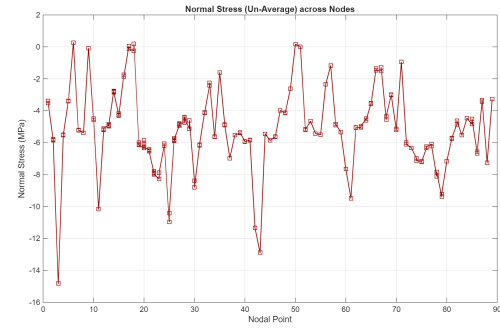


(c) Normal Stress (*Elemental Mean*)

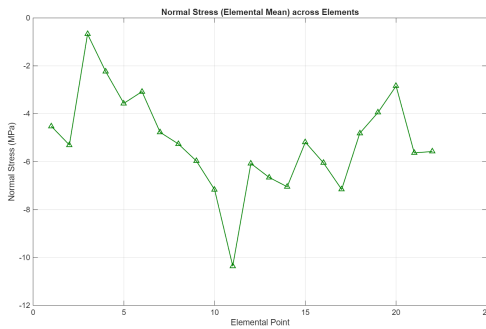
Figure 19: Different Normal Stress in x -axis in ANSYS Simulation.



(a) Normal Stress(Avg) against Nodal Points



(b) Normal Stress(Un-Avg) against Nodal Points



(c) Normal Stress(Elemental Points) against Element

Figure 20: Different Normal Stress Plot against Nodal and Elements.

Comments:

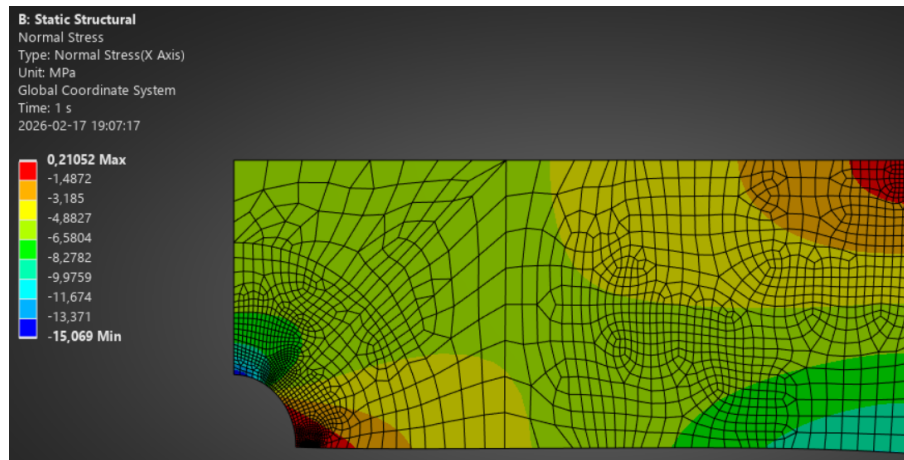
From the plots of averaged , un-averaged and elemental Mean method:

The averaged stress, as shown in Figure 20(a), is a form of post-processing that calculates the average of all the extrapolated nodal values to produce a continuous and smooth contour plot.

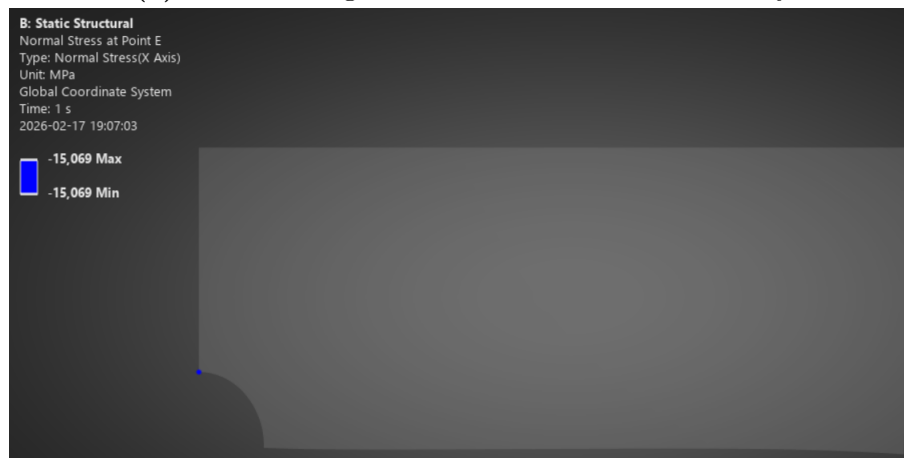
The un-averaged stress (Figure 20(b)) is the most important element for mesh verification. The computation of stress takes place at the integration points (Gauss points), followed by extrapolation to the nodes.

The Elemental Mean (Figure 20(c)) represents the least refined form of the numerical result. Considering that linear elements or a single point value from a quadratic element assume a certain strain field, the Elemental Mean represents the average stress in a particular element volume.

Automatic Mesh Convergence Study

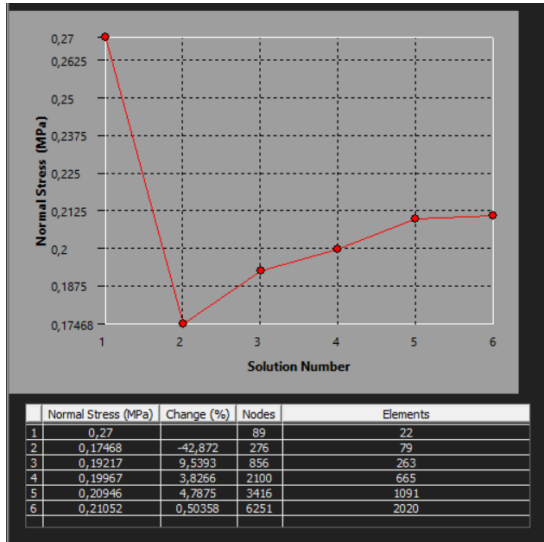


(a) Post Convergence Normal Stress at entire body

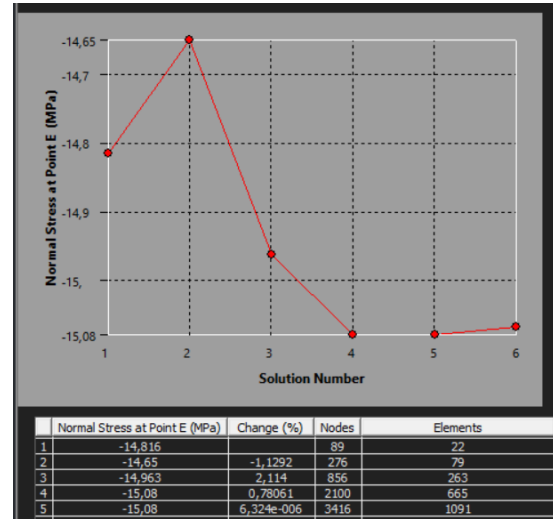


(b) Post Convergence Normal Stress at Point E

Figure 21: ANSYS Simulation Results.



(a) Convergence Plot Normal Stress at Entire Body



(b) Convergence Plot Normal Stress at Point E

Figure 22: Convergence Plot

Results:

The convergence plot shows us that the stress converges at the solution number 6. At this solution number, the normal stress at the point E is 15.06 MPa.

Comments:

The convergence to 15.06 MPa verifies that the numerical model is capable of capturing the physical stress concentration. Before convergence, the normal stress at the point E is 14.816 MPa.

The automatic mesh convergence study results show that the initial mesh was not capable of properly representing the steep stress gradient at the root of the notch, resulting in an initial stress value of 14.816 MPa. Successive adaptive mesh refinement, led to convergence to a stress value of 15.06 MPa.

As shown in Figure 22, coupled with the small difference between successive iterations, confirms that the results have been made mesh-independent, and the converged result provides a sound basis for the computation of the stress concentration factor Kt , ensuring that the model does not underestimate potential failure points.

References

- [1] H Janocha, editor. *Adaptronics and Smart Structures*. Springer-Verlag, 1999.